PREDICTION OF DIABETIC RETINAL DISEASE AND GLAUCOMA AMONG DIABETIC COMMUNITY USING SOFT COMPUTING TECHNIQUES

A THESIS

Submitted by

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This is to certify that no corrections/suggestions were pointed out by the Indian/Foreign Examiners in the Thesis entitled **"Prediction of Diabetic Retinal Disease and Glaucoma Among Diabetic Community Using Soft Computing Techniques"** submitted by **Ms. M. Shanmuga Eswari, (Reg. No. 201906133).**

Place : Krishnankoil
Date : 14.03.2025

Dr. S. BALAMURALI,
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Minutes of the Ph.D. Viva-Voce Examination of Ms. M. Shanmuga Eswari, (Reg. No. 201906133) held on 21st April, 2025 at 11.00 a.m. in Seminar Hall, R & D Office, Kalasalingam Academy of Research and Education.

The Ph. D. Viva-Voce Examination of **Ms. M. Shanmuga Eswari, (Reg. No. 201906133)** on her Ph.D. thesis entitled **"Prediction of Diabetic Retinal Disease and Glaucoma Among Diabetic Community Using Soft Computing Techniques"** was conducted on **21st April, 2025** at **11.00 a.m** in Seminar Hall, R&D Office, Kalasalingam Academy of Research and Education, Anand Nagar, Krishnankoil.

The following members of the Oral Examination Board were present:

1.	**Dr. S. Balamurali** Senior Professor Department of Computer Applications Kalasalingam Academy of Research and Education Krishnankoil – 626 126.	Supervisor & Convener
3.	**Dr. M. Nandhini** Professor of Computer Science School of Engineering and Technology Pondicherry University Puducherry – 605014.	Indian Examiner
4.	**Dr. K. Balasubramanian** Professor and Head Department of Computer Applications Kalasalingam Academy of Research and Education Krishnankoil – 626 126.	DRC Chairman

The research scholar presented the salient features of her Ph.D. work, followed by an interactive session with the board members. During this session, she addressed questions and clarifications raised by both the Indian and Foreign Examiners. The scholar provided comprehensive responses, satisfactorily addressing all queries posed by the board members.

Based on the quality of the scholar's research work, the effectiveness of her presentation, and her satisfactory responses to the examiners' questions, the board unanimously recommends the award of the Ph.D. degree in Computer Applications to Ms. M. Shanmuga Eswari.

Dr. S. Balamurali
(Supervisor & Convener)

21/04/25
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DECLARATION

I do hereby declare that the thesis entitled **“PREDICTION OF DIABETIC RETINAL DISEASE AND GLAUCOMA AMONG DIABETIC COMMUNITY USING SOFT COMPUTING TECHNIQUES”** submitted by me for the Degree of Doctoral in Philosophy in Department of Computer Applications is the result of my original and independent research work carried out under the guidance of **Dr. S. BALAMURALI,** Senior Professor, Department of Computer Applications, Kalasalingam Academy of Research and Education and it has not been submitted for the award of any Degree, Diploma, Associateship, Fellowship of any Universities or Institutions.

Signature of the Candidate
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BONAFIDE CERTIFICATE

It is certified that this doctoral thesis titled **"PREDICTION OF DIABETIC RETINAL DISEASE AND GLAUCOMA AMONG DIABETIC COMMUNITY USING SOFT COMPUTING TECHNIQUES"** is the bonafide work of **Mrs. M. SHANMUGA ESWARI** who carried out the research work under my supervision. It is also certified that to the best of my knowledge the work reported herein does not form a part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other scholar.

Dr. S. BALAMURALI

SUPERVISOR,

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ABSTRACT

Diabetic retinal diseases, including diabetic retinopathy and glaucoma, are irreversible causes of blindness, demanding early detection to mitigate vision loss. This study proposes a novel soft computing-based framework for the prediction of diabetic retinal diseases and glaucoma within diabetic populations, leveraging advanced computational techniques to enhance diagnostic accuracy.

This work introduces a multi-model architecture integrating four innovative soft computing approaches. Firstly, a Random Search Optimization Adaboost Ensemble classifier is designed for diabetic retinal disease and glaucoma prediction, combining adaptive boosting with hyperparameter optimization to address dataset imbalance and improve generalization. Secondly, a Deep Neural Classification Network is developed for diabetic retinal disease detection, employing Region of Interest guided segmentation and Neural Image Optimization, followed by Grey Level Co-occurrence Matrix feature extraction to reduce dimensionality and enhance lesion discrimination. Next, an Artificial Algae Algorithm with Support Vector Machine hybrid model is proposed for glaucoma diagnosis, integrating bio-inspired optimization for feature selection with kernel-based classification to improve boundary delineation in fundus images. Finally, an Interpretable Hybrid Stacking Ensemble Classifier leverages Explainable Artificial Intelligence principles to detect glaucoma in diabetic patients, combining Grey Co-occurrence Matrix and Grey Level Run Length Matrix textural features with meta-learning for transparent decision-making.

The framework is validated using retinal datasets, processed through multilevel preprocessing (noise removal, homomorphic filtering, color normalization) and segmented via Ternausnet and Faster Region-based Convolutional Neural Networks for structural feature extraction. Experimental results demonstrate that the proposed models outperform conventional methods, achieving superior precision 94.3%, recall 92.8%, F1 score 93.5%, and classification accuracy 96.1% with minimized error rates 3.2%. The Random Search Optimization Adaboost Ensemble and Explainable Artificial Intelligence Hybrid Stacking Ensemble Classification models exhibit exceptional interpretability, addressing the "black-box" limitation of deep learning, while the Artificial Algae Algorithm Support Vector

Machine and Deep Neural Classification Network optimize computational efficiency and specificity.

This study highlights the transformative potential of soft computing techniques ensemble learning, deep feature engineering, bio-inspired optimization, and explainable artificial intelligence in advancing early diagnosis of diabetic eye diseases. By harmonizing accuracy, interpretability, and clinical relevance, the proposed framework offers a scalable solution for reducing preventable blindness in diabetic communities, aligning with global healthcare priorities. This research underscores the transformative role of artificial intelligence in advancing health justice, ensuring that diabetic communities regardless of geography or socioeconomic status can access timely, life changing interventions

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M. SHANMUGA ESWARI

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LIST OF SYMBOLS AND ABBREVIATIONS

SYMBOLS	ABBREVIATION
A/V	Artery/Vein
AAA	Artificial Algae Algorithm
AAASVM	Artificial Algae Algorithm with Support Vector Machine
AI	Artificial Intelligence
AMD	Age-Related Macular Degeneration
ANN	Artificial Neural Network
AUC	Area Under Curve
BOSVM	Bayesian Optimization Support Vector Machine
CAD	Computer Aided Design
CAV	Convolution Along Vessel
CDR	Cup Disc Ratio
CNN	Convolutional Neural Networks
CNN-LSTM	CNN-Long Short-Term Memory
CoG-NET	Classification of Glaucoma Network
CWS	Cotton Wool Spots
CycleGAN	Cycle Generative Adversarial Network
DCNN	Deep Convolutional Neural Network
DED	Diabetic Eye Disease
DeiT	Data-Efficient Image Transformer
DFP	Digital Fundus Photography
DL	Deep Learning
DM	Diabetic Mellitus
DME	Diabetic Macular Edema
DNCN	Deep Neural Classification Network
DNN	Deep Neural Network
DOMLS	Deep Optimized Machine Learning Strategy
DPA	Depthwise Parallel Attention

DR	Diabetic Retinopathy or Diabetic Retinal Disease
DRIVE	Digital Retinal Image Vessel Extraction
DS	Decision Stumps
DT	Decision Tree
EAIHSEC	Explainable Artificial Intelligence Hybrid Stacking Ensemble Classification
ECNet	Evolutionary Convolutional Network
EX	Exudates
FANet	Fully Attention based Network
FJAM	Fuzzy Joint Attention Module
FN	False Negative
FP	False Positive
FRCNN	Faster Region Based Convolutional Neural Network
FRCNNAAASVM	Faster Region-based Convolutional Neural Network- Artificial Algae Algorithm Support Vector Machine
GLCOM	Grey Level Co-occurrence Matrix
GLRLM	Grey Level Run Length Matrix
GON	Glaucomatous Optic Nerve
HEM	Hemorrhage
HRF	High Resolution Fundus
HSEC	Hybrid Stacking Ensemble Classifier
IoP	Intraocular Pressure
LBP	Local Binary Patterns
LQ	Low Quality
MA	Microaneurysms
MCG-Net	Multilabel Convolutional Graph Network
MKMOL	Modified K-Means Optimization Logic
ML	Machine Learning
MLP	Multi Layer Perceptron
MSE	Mean Square Error
NIO	Neural Image Optimization

NN	Neural Network
OAL	Ocular Axial Length
OC	Optical Cup
OCT	Optical Coherence Tomography
OD	Optical Disc
OHTS	Ocular Hypertension Treatment Study
ONH	Optical Nerve Head
OWDANN	Optimal Weighted Based Deep Artificial Neural Network
POAG	Primary Open Angle Glaucoma
RBF	Radial Basis Function
RDR	Rim Disc Ratio
ReLU	Rectified Linear Unit
RF	Random Forest
RMSE	Root Mean Square Error
RNFL	Retinal Nerve Fiber Layer
ROC	Receiver Operating Characteristic
ROI	Region of Interest
RSOAE	Random Search Optimization Adaboost Ensemble
SAP	Standard Automated Perimetry
SLO	Scanning Laser Ophthalmoscopy
SMO	Sequential Minimal Optimization
SOA	Swarm Optimization Algorithm
SURF	Speeded Up Robust Features
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
VCDR	Vertical Cup Disc Ratio
ViT	Vision Transformer
XAI	Explainable Artificial Intelligence

SYMBOLS

i	Hyperparameter
$f(i)$	Function
Θ	Parameter Space
x_a, y_a	Variable Ranges
s	Samples
$g_z(i)$	Weak Classifier
ε_z	Weighted Classification Error
x, y	Image Intensity
H	Feature Map
k	Layer
C	Convolution Kernal
b	Bias
σ	Activation Function
$\mathcal{L}$	Loss Function
η	Learning Rate
$\hat{S}$	Predicted Segment S
S	Ground Truth
$\nabla_\theta \mathcal{L}$	Loss Gradient
W	Weight
β	Hyper Parameter
$x_i x_j$	Kernal Input Samples
γ	Positive Scalar
Y_i	Algae Population
NP	Total Number of Algae in Population
D	Size of Algae colony
r_j	Number of Classifiers
D	Distance
P	Cooccurence Matrix
$\Delta x, \Delta y$	Offset of Neighbour Pixel

gl	Grey Level
rl	Run Length
Mg	Maximum Gray
$Smax$	Maximum Length
$\bar{A}$	Standard Mean
h_i	Dataset
X_{i}'	Feature Vector
$\mathrm{H}(\mathrm{x})$	Hybrid Classifier